Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	01 November 2026
Team ID	[Enter Team ID]
Project Name	Global Olympic Sports Performance Analysis Using Tableau
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
FR-1	Data Access & Integration	Connect the Olympic athlete-events dataset (athlete_events.csv) containing athlete, NOC, year, season, sport, event and medal fields. Integrate supplementary country-region, host-country and sports-investment tables when obtained.	High
FR-2	Data Processing	Check data types, handle null medal values, standardize country/NOC and sport labels, filter the analysis period, distinguish Summer and Winter Games, and create a distinct medal key to avoid counting every team member as a separate medal.	High
FR-3	Business Analysis	Calculate distinct medal totals by country, sport, year and host nation; calculate athlete participation and medals-per-athlete measures; compare regional efficiency; and analyze investment against medal outcomes when reliable investment data is available.	High
FR-4	Business Insights & Analytical Outputs	Provide nine visualizations: sport-wise medals per athlete; country medal ranking; sport medal distribution; investment-versus-medal scatter/bubble chart; regional efficiency chart; country efficiency ranking; medal trend line; host-nation comparison; and regional medal distribution by sport. Organize them into three dashboards and a four-scene Tableau story.	High
FR-5	User Interaction & Analysis	Allow users to filter by year, season, NOC/ country, sport, medal type and region where the corresponding data exists. Enable dashboard actions/tooltips so selecting a mark helps explore related views.	Medium
FR-6	Data Update & Reporting	Allow refreshed CSV/Excel source files to be reconnected or refreshed in Tableau. Provide dashboards, story, project documentation and an end-to-end demonstration video; publish/embed the dashboards through Tableau and Flask if deployment access is available.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboards should have clear titles, readable labels, consistent colours, simple navigation, useful tooltips and easy-to-use filters.
NFR-2	Security	Use public/non-sensitive Olympic data. Do not expose credentials or private configuration in the Flask application; use authorized Tableau sharing and embedding settings.
NFR-3	Reliability	Calculated fields and medal totals should be validated against source records. Refreshes should not silently change field definitions or duplicate records.
NFR-4	Performance	Use only required fields, avoid unnecessary marks, optimize filters and test dashboard load time and interaction responsiveness.
NFR-5	Availability	Published dashboards should be accessible to intended viewers when the Tableau hosting service and network are available. Local Tableau files remain available for offline development.
NFR-6	Scalability	Design the data source with separate lookup tables and reusable calculations so additional Olympic years, countries, sports and investment records can be added.